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BASIC MATHEMATICS

FORM FOUR

Certificate of Secondary Education Examination (CSEE) 2025 Answer Key and Worked Solutions

Subject	Basic Mathematics
Code	041
Level	Form Four
Year	2025
Exam Board	NECTA
Exam	Certificate of Secondary Education Examination (CSEE)
Type	Answer key and worked solutions
Questions	14 compulsory questions; 33 answered parts

Try each question before checking the solution. Follow the steps, compare your working, and practise the parts you find difficult.

Source paper: BasicMath-F4-2025.pdf

Question 12(b)(ii) includes the complete outside area of the closed box and identifies the alternative lateral-area reading of “sides”.

Section A — Worked solutions

1(a). Express 0.11363636... as a fraction in its simplest form; only 36 recurs.

Step 1: Let $p = 0.11363636...$. Moving two and four decimal places gives $100p = 11.363636...$ and $10,000p = 1136.363636...$.

Step 2: Subtract to cancel the recurring tail: $9900p = 1125$. Thus $p = 1125/9900 = 5/44$.

Final Answer: 5/44.

Study Tip: The two initial 1s do not repeat. Keep the recurring block aligned before subtracting.

1(b)(i). Write 0.00070482 to three significant figures.

Step 1: Start counting significant digits at the first nonzero digit: 7,0,4. The following digit is 8.

Step 2: Round 704 up to 705, keeping the original decimal position.

Final Answer: 0.000705, or 7.05×10^{-4} .

1(b)(ii). Round 1,233,388 to the nearest ten thousand.

Step 1: The ten-thousands digit is 3; the following thousands digit is 3.

Step 2: Since the following digit is below 5, retain the ten-thousands digit and replace all smaller places with zero.

Final Answer: 1,230,000.

1(c). Three-fifths of Tanzania's people live in cities in the stated model. What percentage does not live in cities?

Step 1: The complement is $1 - 3/5 = 2/5$.

Step 2: Convert to a percentage: $(2/5) \times 100 = 40\%$.

Final Answer: 40%.

2(a). Express $27^n \times 9^{2n} \times 3$ as a single power of 3.

Step 1: $27^n = (3^3)^n = 3^{3n}$ and $9^{2n} = (3^2)^{2n} = 3^{4n}$.

Step 2: When multiplying powers with the same base, add exponents: $3n + 4n + 1 = 7n + 1$.

Final Answer: 3^{7n+1} .

2(b). Rationalise the denominator of $(\sqrt{2} + 1)/(\sqrt{2} - \sqrt{5})$.

Step 1: Multiply numerator and denominator by the conjugate $\sqrt{2} + \sqrt{5}$.

Step 2: The denominator becomes $2 - 5 = -3$. Expanding the numerator gives $2 + \sqrt{10} + \sqrt{2} + \sqrt{5}$. Divide by -3 .

Final Answer: $-(2 + \sqrt{10} + \sqrt{2} + \sqrt{5})/3$.

2(c). Given $\log_{10}2 = 0.3010$ and $\log_{10}3 = 0.4771$, find $\log_{10}72$.

Step 1: Factor $72 = 2^3 \times 3^2$. Apply the product and power rules: $\log 72 = 3\log 2 + 2\log 3$.

Step 2: Calculate $3(0.3010) + 2(0.4771) = 0.9030 + 0.9542 = 1.8572$.

Final Answer: 1.8572.

3(a)(i). Of 260 students, 130 take Physics, 150 Chemistry and 40 both. How many take Physics only?

Step 1: The Physics total includes those taking both subjects.

Step 2: Subtract the overlap: $130 - 40 = 90$.

Final Answer: 90 students.

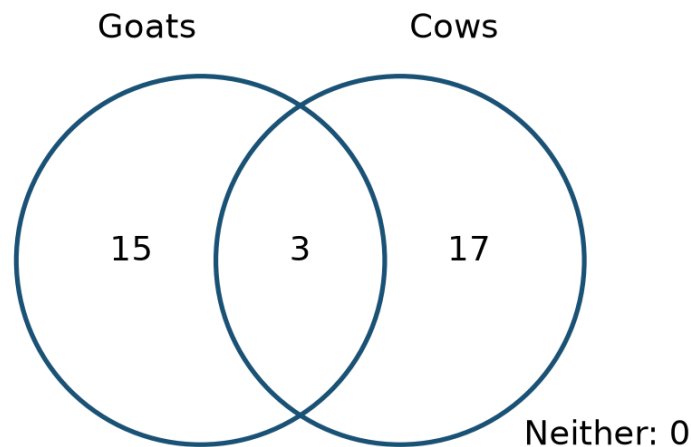
3(a)(ii). How many of those students take neither subject?

Step 1: Union = $130 + 150 - 40 = 240$. The overlap is subtracted because it was counted twice.

Step 2: Neither = total – union = $260 - 240 = 20$.

Final Answer: 20 students.

3(b). Among 35 livestock keepers, 18 keep goats, 20 cows and 3 both. Draw a Venn diagram and find $P(\text{goats only})$.



Step 1: Place 3 in the intersection. Goats only = $18 - 3 = 15$; cows only = $20 - 3 = 17$.

Step 2: Union = $15 + 3 + 17 = 35$, so none lie outside both circles. Favourable keepers = 15; probability = $15/35 = 3/7$.

Final Answer: Goats only 15; both 3; cows only 17; neither 0. $P(\text{goats only}) = 3/7$.

4(a). In triangle ABC, $AB = 8$ cm, exterior angle $ABD = 150^\circ$ and area = 22.6 cm^2 . Find BC.

Step 1: Interior $\angle ABC = 180^\circ - 150^\circ = 30^\circ$. Use area = $\frac{1}{2} AB \times BC \times \sin \angle ABC$.

Step 2: $22.6 = \frac{1}{2} \times 8 \times BC \times \frac{1}{2} = 2BC$. Therefore $BC = 22.6/2 = 11.3$ cm.

Final Answer: 11.3 cm.

4(b). Find the perimeter of a regular pentagon inscribed in a circle of radius 10 m.

Step 1: Each central angle is $360^\circ/5 = 72^\circ$. Bisecting one isosceles triangle gives a right triangle with angle 36° and hypotenuse 10 m.

Step 2: Half a side = $10\sin 36^\circ$, so a side = $20\sin 36^\circ$. Multiply by five sides: perimeter = $100\sin 36^\circ$.

Final Answer: $100 \sin 36^\circ \approx 58.78$ m.

5(a). Using the stated exchange rate Tsh 2300 per US dollar, convert Tsh 27,600,000 to dollars.

Step 1: One dollar costs Tsh 2300, so divide the shilling amount by 2300.

Step 2: $27,600,000 \div 2300 = 12,000$. Check: $12,000 \times 2300 = 27,600,000$.

Final Answer: US \$12,000.

5(b). Cylinder height varies directly as volume and inversely as radius squared. Height 3 cm corresponds to volume 900 cm^3 and radius 10 cm. Find height for 1200 cm^3 and radius 5 cm.

Step 1: Write $h = kV/r^2$. Substitute the first conditions: $3 = k \times 900/100 = 9k$, so $k = 1/3$.

Step 2: For the new cylinder, $h = (1/3)(1200)/25 = 400/25 = 16 \text{ cm}$.

Final Answer: 16 cm.

6(a). For $a = 2i - j$, $b = -i + 3j$ and $c = 3i - 4j$, find the magnitude of $a + b + c$.

Step 1: Add corresponding components: $a + b + c = (2 - 1 + 3)i + (-1 + 3 - 4)j = 4i - 2j$.

Step 2: Magnitude $= \sqrt{4^2 + (-2)^2} = \sqrt{20} = 2\sqrt{5}$.

Final Answer: $2\sqrt{5}$, approximately 4.47.

6(b). A line segment has midpoint (3,5) and one endpoint (6,7). Find the other endpoint.

Step 1: Let the missing endpoint be (x,y) . The midpoint formula gives $(6 + x)/2 = 3$ and $(7 + y)/2 = 5$.

Step 2: Thus $x = 6 - 6 = 0$ and $y = 10 - 7 = 3$. Check the averages are (3,5).

Final Answer: (0,3).

7(a). Sand, gravel and cement are mixed in ratio 500:600:200. How much cement is in 39,000 kg of mixture?

Step 1: Total ratio parts $= 500 + 600 + 200 = 1300$. Cement contributes $200/1300$ of the mixture.

Step 2: Cement mass $= (200/1300) \times 39,000 = 200 \times 30 = 6000 \text{ kg}$.

Final Answer: 6000 kg of cement.

7(b). Opening stock Tsh 2000, purchases 13,000, closing stock 500, sales 30,000 and expenses 6500: find net profit.

Step 1: Cost of goods sold = opening stock + purchases – closing stock = $2000 + 13,000 - 500 = 14,500$.

Step 2: Gross profit = sales – cost of goods sold = $30,000 - 14,500 = 15,500$.

Step 3: Net profit = gross profit – expenses = $15,500 - 6500 = 9000$.

Final Answer: Tsh 9000.

8(a). Calculate the amount after investing Tsh 10,000 at 5% compound interest per year for three years.

Step 1: For annual compounding, $A = P(1 + r)^n$ with $P = 10,000$, $r = 0.05$ and $n = 3$.

Step 2: $A = 10,000 \times 1.05^3 = 10,000 \times 1.157625 = 11,576.25$. The interest earned is Tsh 1576.25.

Final Answer: Tsh 11,576.25.

8(b). The numbers 3, $x - 1$ and 27 form a geometric progression. Find x .

Step 1: For consecutive GP terms, the middle term squared equals first $\times$ third: $(x - 1)^2 = 3 \times 27 = 81$.

Step 2: $x - 1 = \pm 9$, so $x = 10$ or -8 . Check the sequences 3, 9, 27 and 3, -9 , 27 have ratios 3 and -3 respectively.

Final Answer: $x = 10$ or $x = -8$.

Study Tip: A geometric progression can have a negative common ratio. Retain both solutions unless a positive ratio is required.

9(a). Find the largest angle in a triangle with side lengths 6, 7 and 8 cm.

Step 1: The largest angle is opposite the longest side, 8 cm. Apply the cosine rule: $8^2 = 6^2 + 7^2 - 2(6)(7)\cos\theta$.

Step 2: $\cos\theta = (36 + 49 - 64)/84 = 21/84 = 1/4$. Thus $\theta = \cos^{-1}(1/4) \approx 75.52^\circ$.

Final Answer: Approximately 75.52° .

9(b). ABC is right-angled at B, with AC = 6 cm and AB = 3 cm. Find $\tan \angle C$.

Step 1: AC is the hypotenuse. By Pythagoras, $BC = \sqrt{6^2 - 3^2} = \sqrt{27} = 3\sqrt{3}$.

Step 2: Relative to angle C, the opposite side is AB and the adjacent side is BC. Therefore $\tan C = 3/(3\sqrt{3}) = 1/\sqrt{3}$.

Final Answer: $1/\sqrt{3}$, or $\sqrt{3}/3$.

10(a). Given $x = 1/t^5$, rewrite $2/t^{10} - 3/t^5 + 1 = 0$ in terms of x .

Step 1: Since $x = t^{-5}$, $x^2 = t^{-10}$. The original equation requires $t \neq 0$.

Step 2: Replace $1/t^{10}$ with x^2 and $1/t^5$ with x .

Final Answer: $2x^2 - 3x + 1 = 0$.

10(b). Solve the resulting quadratic $2x^2 - 3x + 1 = 0$.

Step 1: Factor: $2x^2 - 3x + 1 = (2x - 1)(x - 1)$.

Step 2: Set each factor to zero: $2x - 1 = 0$ gives $x = 1/2$; $x - 1 = 0$ gives $x = 1$. If t is also needed, $t = 1$ or $\sqrt[5]{2}$.

Final Answer: $x = 1$ or $x = 1/2$.

Section Summary: Align recurring decimals before subtracting. Complements and Venn diagrams avoid double counting; ratios and variation equations must use consistent units.

Section B — Worked solutions

11(a). Marks 65-69, 70-74, 75-79, 80-84, 85-89, 90-94 have frequencies 10, 12, 55, 10, 5, 8. How many pupils scored 75-79?

Step 1: Find the class interval 75-79 in the table.

Step 2: Its frequency is 55. Frequencies count pupils; do not use the class midpoint or cumulative total.

Final Answer: 55 pupils.

11(b). Calculate the grouped median of the marks to the nearest whole number.

Step 1: Total $N = 100$; cumulative frequencies are 10,22,77,87,92,100. $N/2 = 50$ lies in the 75–79 class.

Step 2: Use continuous lower boundary $L = 74.5$, preceding cumulative frequency 22, class frequency 55 and width 5.

Step 3: Median $= 74.5 + [(50 - 22)/55] \times 5 = 77.04545\dots$. Round to the nearest whole mark: 77.

Final Answer: 77 marks.

12(a). Find the distance between X(60°N,30°E) and Y(60°N,40°E), taking a degree at the equator as 111.6 km.

Step 1: Longitude difference $= 40^\circ - 30^\circ = 10^\circ$. At latitude 60° , the parallel's radius is the equatorial radius multiplied by $\cos 60^\circ = 1/2$.

Step 2: Distance along that parallel $= 10 \times 111.6 \times 1/2 = 558$ km.

Final Answer: 558 km along the parallel of latitude.

Study Tip: This is the parallel-arc distance requested by the latitude model, rather than a great-circle route.

12(b)(i). A closed rectangular box has dimensions 40 cm by 30 cm by 50 cm. Find its space diagonal AG.

Step 1: The base diagonal squared is $40^2 + 30^2 = 2500$.

Step 2: The space diagonal squared adds the height squared: $AG^2 = 2500 + 50^2 = 5000$. Therefore $AG = \sqrt{5000} = 50\sqrt{2}$.

Final Answer: $50\sqrt{2}$ cm, approximately 70.71 cm.

12(b)(ii). Find the outside surface area of the closed box.

Step 1: There are two faces of each size: 40×30 , 40×50 and 30×50 .

Step 2: Total $= 2(1200 + 2000 + 1500) = 9400$ cm².

Final Answer: 9400 cm² for all six outer faces.

Study Tip: If “the sides” is restricted to the four vertical faces, their lateral area is $2(40 + 30) \times 50 = 7000$ cm². The complete closed-box outside area includes its top and bottom.

13(a). $G = \begin{bmatrix} -1 & 11 \\ a & 3 \end{bmatrix}$ is singular. Find a .

Step 1: A singular square matrix has determinant zero. $\det G = (-1)(3) - 11a = -3 - 11a$.

Step 2: Set $-3 - 11a = 0$, giving $a = -3/11$.

Final Answer: $a = -3/11$.

13(b). Use a matrix method to solve $5x + 3y = 9$ and $10x + 7y = 11$.

Step 1: Write $A = \begin{bmatrix} 5 & 3 \\ 10 & 7 \end{bmatrix}$, $u = [x, y]^T$ and $b = [9, 11]^T$, so $Au = b$. $\det A = 35 - 30 = 5$.

Step 2: $A^{-1} = (1/5)\begin{bmatrix} 7 & -3 \\ -10 & 5 \end{bmatrix}$. Multiply by b : $u = (1/5)[63 - 33, -90 + 55]^T = [6, -7]^T$.

Step 3: Check: $5(6) + 3(-7) = 9$ and $10(6) + 7(-7) = 11$.

Final Answer: $x = 6, y = -7$.

14(a). Given $f(x) = x + 2$ and $g(x) = 3x - 2$, find the product of $f^{-1}(x)$ and $g^{-1}(x)$.

Step 1: To invert f , write $y = x + 2$ and solve $x = y - 2$: $f^{-1}(x) = x - 2$. To invert g , $y = 3x - 2$ gives $g^{-1}(x) = (x + 2)/3$.

Step 2: Multiply the inverse values: $(x - 2)(x + 2)/3 = (x^2 - 4)/3$.

Final Answer: $f^{-1}(x)g^{-1}(x) = (x^2 - 4)/3$.

Study Tip: The paper asks for the product of the inverse functions. A product multiplies their values; a composition would substitute one function into the other.

14(b). A tailor has 150 m² cotton and 90 m² wool. A suit needs 1 m² cotton and 2 m² wool; a gown needs 3 m² cotton and 1 m² wool. Revenues are Tsh 40,000 per suit and 60,000 per gown. Formulate the revenue-maximising problem.

Step 1: Let x be the number of suits and y the number of gowns. Cotton used = $x + 3y$ and wool used = $2x + y$.

Step 2: Usage cannot exceed available cloth, so both constraints use $\leq$. Multiply each garment count by its revenue to form the objective. Nonnegative whole numbers reflect garment counts.

Final Answer: Maximise $Z = 40,000x + 60,000y$, subject to $x + 3y \leq 150$; $2x + y \leq 90$; $x, y \geq 0$, with x, y whole numbers.

Section Summary: Use class boundaries for grouped medians. A space diagonal includes all three squared dimensions. Inverse functions, products and matrix inverses are different operations.